

All-in-One Hybrid Inverter — Quick Start

KR-L8K-SPUS · KR-L10K-SPUS · **KR-L12K-SPUS** · KR-L15K-SPUS | 120/240 V split phase, 60 Hz

KOOR

Keep this sheet with the inverter. It covers the steps you use on the job. Full detail, settings and fault codes are in the **User Manual** in this same book.
Qualified electrical personnel only.

1 WHAT IS ON THE UNIT

Front panel — indicator LEDs	Outside of the unit	Inside the wiring compartment
1 Green — ON: inverter is running. Flashing: standby.	9 PV DC disconnect	18 Battery terminals
2 Blue — ON: communication with the battery BMS is normal.	10 LCD screen	19 Loads
3 Yellow — ON: inverter is in EPS (backup) mode.	11 BAT breaker	20 3 × PV inputs
4 Red — inverter is in fault mode.	12 Parallel RJ45 ports	21 Generator terminals
Buttons	13 Input pinouts for sensors & accessories	22 Load terminals
	14 BMS RJ45 ports (RS485 / CAN)	23 Ground busbar
	15 Load breaker	24 Grid terminals
	16 Wi-Fi dongle port	
	17 ON / OFF button	

Parallel systems

Up to **6 units** in parallel — up to **90 kW**. All units must be on the **same software version** (check MCU numbers under System Setup). A parallel system **requires one joint battery bank**, and all inputs and outputs must be shared across every inverter.

2 WHERE IT CAN HANG

Site limits	Yes	No
Max ambient temp	+60 °C	• No tilting forward, backward or laying it down
Min ambient temp	–25 °C	• No water pipes or cables buried in the wall behind it
Relative humidity	up to 100% (no condensation)	• No flammable material, no corrosive or explosive area
Altitude	up to 4000 m	• No direct sun, rain or snow on the unit
Enclosure	High sealed design, NEMA 4X	
Clearance	500 mm free on every side	

3 MOUNTING — 5 STEPS

1 Mark the bracket	2 Drill 10 mm holes	3 Fasten the bracket to the wall	4 Lift the inverter onto the bracket	5 Lock it down with the M8 anti-collision bolts
--------------------	----------------------------	----------------------------------	--------------------------------------	---

4 WHAT IS IN THE BOX

Item	Qty	Description	Item	Qty	Description
A	1	Hybrid inverter	G	1	User manual
B	10	Tubular terminal (20 AWG)	H	1	Wi-Fi plug (dongle)
C	4	Stainless steel anti-collision bolt M8	I	2	Sensor clamp (CT)
D	1	Communication cable	J	10	Tubular terminal (6 AWG)
E	2	Copper terminal (1/0 AWG)	K	1	Mounting bracket
F	12	Tubular terminal (10 AWG)	Check the box and the unit for damage before you sign for it. Note any dents, holes or cracks on the freight paperwork.		

Wiring, Start-Up and Checks

Koor All-in-One KR-L8K / L10K / **L12K** / L15K-SPUS | Quick Start page 2

KOOR

5 CABLES

#	Cable	Type	Size
1	Grounding	Single multi-core yellow/green copper	10 AWG or bigger
2	AC output	Two or three different color multi-core copper	2/0 AWG or bigger
3	PV input	PV cable (such as PV1-F)	12 AWG – 10 AWG
4	Battery	Red and black multi-core copper	2/0 – 4/0 AWG
5	Communication	CAT5E	—

Strip lengths

Generator conductors (6 – 4 AWG)	16.5 mm
Grid, load and battery conductors	16.5 mm
PV conductors (12 – 10 AWG)	16 mm

Before you touch a terminal

- **Every switch OFF** — PV, battery, grid, load, generator. Never work on it with power on.
- The lithium battery has its **own** power line and comm line — use the ones that came with it.
- The inverter ships with the basic comm line — use it directly.
- **Do not reverse battery polarity.** It damages the system and voids the warranty.
- Never land the grid on the **LOAD** output terminal.

PV check before you land it

Meter every string open-circuit before connecting: $\leq$

525 V

PV+ = RED · PV- = BLACK. Crimp the pin, push the cable in firmly and press the catch until it clicks.

6 COMMUNICATION PORTS

	Port	What it is for
A	NTC	Temperature sensor for a lead-acid battery
B	CT interface	Current sensor clamps
C	GEN (NO)	Generator auto-start dry contact. When the GEN signal is active the contact closes — no voltage output.
D	± 12 V RSD	Power for the RSD PLC transmitter (500 mA max)
E	EPO	Reserved for an external emergency stop
F	Meter RS485	Energy meter communication
G	ATS	240 V output port, live when the inverter is on
H	Battery comm	CAN & RS485 to the battery BMS
I	Parallel comm	CAN & RS485 between inverters
J	Parallel CAN resistor	DIP switch — set it when inverters run in parallel

BMS RJ45 pinout (battery port)

Pin	Signal
1	BAT RS485 A
2	BAT RS485 B
3	NC
4	BAT CAN H
5	BAT CAN L
6	NC
7	BAT RS485 B
8	BAT RS485 A

Only the **master** battery gets a comm cable to the inverter. Set battery type and brand on the screen — comms come up in 1–2 minutes. Wrong protocol is not covered by warranty.

7 TURN IT ON — IN THIS ORDER

- 1 Press and hold the **ON/OFF** button to turn the unit on.
- 2 Turn on the **PV** switch.
- 3 Turn on the switch between the **grid** and the inverter.
- 4 Turn on the switch between the **battery** and the inverter, and **wake the battery up**.
- 5 Set the inverter up — see the User Manual for every setting.

8 SHUT IT DOWN

- 1 Battery switch off.
- 2 Grid switch off.
- 3 PV switch off.
- 4 Hold the **ON/OFF** button to turn the unit off.

Shutdown is start-up backwards. Before power on, make sure every voltage and current is inside the inverter spec — outside it, the inverter gets damaged.

9 POST-INSTALL CHECK

☐ Inverter mounted correctly and firmly	☐ All switches off
☐ Wi-Fi / BLE stick installed correctly and firmly	☐ Ground wire connection confirmed and reliable
☐ Cable runs reasonable, no broken insulation	☐ All wires correct and securely connected
☐ Cable ties trimmed, no sharp corners	☐ Exposed terminals protected, no vacant ports
☐ All packing and scrap material cleaned up and hauled out	

10 APP AND MONITORING

Web	home.solarmanpv.com
iPhone	App Store — search “ Solarman Smart ”
Android	Google Play — search “ Solarman Smart ”

Wi-Fi stick lights. COM off + READY slow flash = not talking to the inverter; reset it, hold Reset 5 s. NET off + COM on = never joined the router. Fast flashing is normal. Full table: **User Manual 8.5.**